

On the Power of Mediation in Games

Fedor Sandomirskiy Princeton University

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based on new results and on joint work with

[Kirill Rudov](#) & [Leeat Yariv](#) (*Extreme Equilibria*)

[Ben Wincelberg](#) (*Delegation in Games and Equilibrium Uniqueness*)

The question

- NE: independent, decentralized play
- CE (Aumann, 1974): a mediator sends players possibly correlated recommendations
- multiple NE already make mediation useful for coordination
- when is correlation only equilibrium selection, and when does it do more?

When is $\text{conv}(\text{NE}) = \text{CE}$ and when are they different?

A menu of motivations

The gap $CE(G) \setminus \text{conv}(\text{NE}(G))$ is where correlation has **genuine bite**:

- **Design / recommendations:** can a recommender sustain outcomes beyond decentralized play?
- **Communication:** is pre-play talk just coordination on which NE to play?
(Bárány, 1992; Forges, 1990; Lehrer and Sorin, 1997; Ben-Porath, 1998; Gerardi, 2004)
- **Learning:** can no-regret / calibrated dynamics find decentralized prediction?
(Foster and Vohra, 1997; Fudenberg and Levine, 1999; Hart and Mas-Colell, 2000)
- **Computation:** when does a tractable LP describe a decentralized prediction?
(Papadimitriou and Roughgarden, 2008)
- **Decision theory:** when does Bayesian rationality collapse to Nash logic?
(Aumann, 1987)

What is known

CE can be strictly larger than $\text{conv}(\text{NE})$:

- the original 2×2 example ([Aumann, 1974](#)); all non-degenerate 2×2 games with two pure NE ([Calvó-Armengol, 2006](#)); mediation value in 2×2 games ([Ashlagi, Monderer, and Tennenholtz, 2008](#))

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Extremality of a NE inside CE:

- always extreme in two-player games ([Cripps, 1995](#); [Evangelista and Raghavan, 1996](#); [Canovas, Hansen, and Jaumard, 1999](#)); non-extreme iff 3+ players mix ([Rudov, Sandomirskiy, and Yariv, 2025](#))

Outline

- ① Model: games, CE, NE, regularity
- ② Characterization of games with $\text{CE}(G) \setminus \text{conv}(\text{NE}(G)) \neq \emptyset$
- ③ How to construct points outside the hull
 - ▶ Tools: extremality and index theory
- ④ Games with a unique CE and CCE

Model

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Correlated equilibrium (Aumann, 1974)

$\mu \in \Delta(A)$ is a CE if every recommendation is self-enforcing: for all i and all $a_i, a'_i \in A_i$

$$\sum_{a_{-i}} \mu(a_i, a_{-i}) [u_i(a_i, a_{-i}) - u_i(a'_i, a_{-i})] \geq 0$$

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- $\text{CE}(G)$ is a **polytope**: finitely many linear inequalities on $\Delta(A)$
- an NE is a CE that is a **product**: $\mu = \mu_1 \times \dots \times \mu_n$
- $\text{conv}(\text{NE})$ = what a mediator reaches by only coordinating which NE to play

We need a non-degeneracy assumption

- pathological games blur the question
- if $u_i \equiv 0$ for all i , then $\text{CE} = \text{conv}(\text{NE}) = \Delta(A)$
- correlation adds nothing **for the wrong reason**: there are too many NE

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G is **regular** if it has finitely many NE and each is

- **quasi-strict**: incentive constraints are slack outside of the support
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- each NE is then stable: it persists, and moves smoothly, under small payoff perturbations
 - **Harsanyi (1973)**: all games except a measure-zero set are regular

Characterization

Main result: a sharp dichotomy

Theorem

Let G be regular. Then exactly one of the following holds:

$$\underbrace{CE \setminus \text{conv}(NE) \neq \emptyset}_{\text{mediation has real bite}}$$

OR

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Interpretation

- communication and recommendation systems expand the set of decentralized outcomes whenever coordination failure is possible
- learning dynamics and LP solving are guaranteed to reach a decentralized outcome only in maximally robust games

Extension to Bayesian games

States Θ , finite types T_i , prior $\pi \in \Delta(\Theta \times T)$, payoffs $u_i(a, \theta, t)$

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- $\text{CE (mediator knows nothing)} \subseteq \text{communication equilibria (players report types)} \subseteq \text{BCE (mediator omnipotent)}$
- $\Rightarrow$ the same dichotomy holds for communication equilibria and BCE

How to construct points outside the hull

Where do points outside the hull come from?

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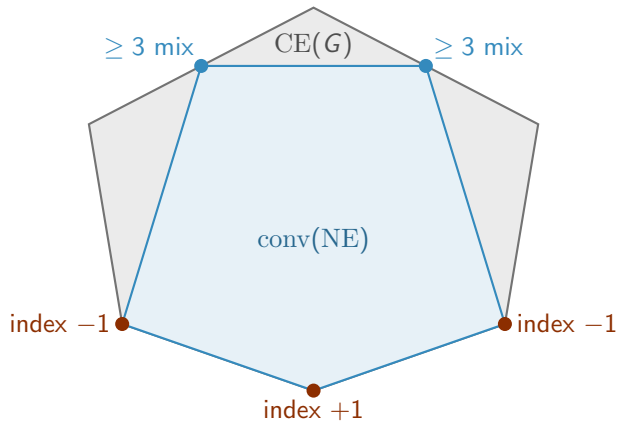
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Two regimes require different arguments:

some NE has ≥ 3 mixers:	extremality-based argument
every NE has ≤ 2 mixers:	index-theory-based argument

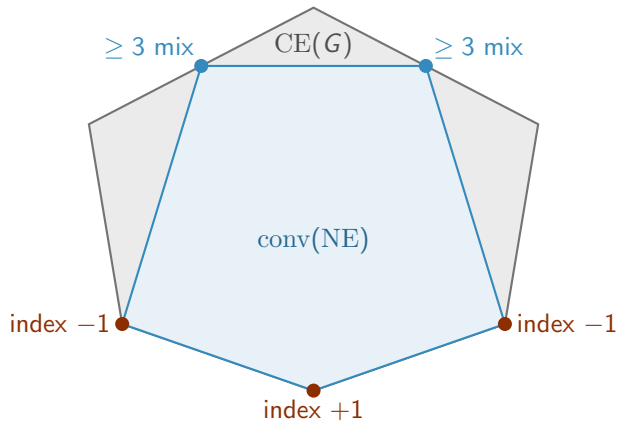
The two ways to leave the hull



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- NE with ≥ 3 mixers sit inside faces: corners of CE escape the hull
- NE with ≤ 2 mixers sit at **vertices** of CE (extreme)
 - ▶ $\text{index} -1 \Rightarrow$ admit a perturbation exiting the hull

Equilibria with three or more mixing players

- [Rudov, Sandomirskiy, and Yariv \(2025\)](#): A quasi-strict NE with ≥ 3 mixing players is **not extreme**
- **Rough idea:**
the dimension of perturbations of ν exceeds the number of incentive constraints

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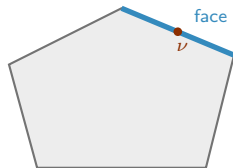
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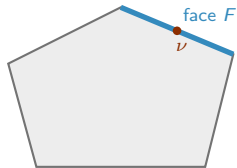
$\Rightarrow$ those extreme points lie **outside** $\text{conv}(\text{NE})$



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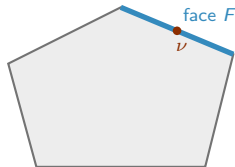
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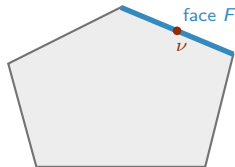
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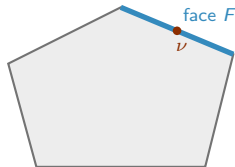
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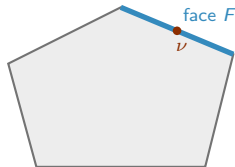
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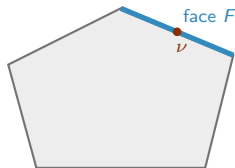
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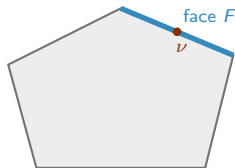
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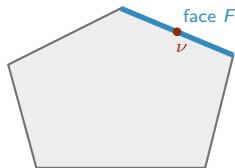
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- so e also has ≥ 3 mixers, contradicting extremality



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Two ideas to the rescue

- 1 **anti-Nash**: equilibria of the auxiliary game with $u'_i = -u_i$ can help construct incentive-compatible perturbations of an NE in the original game (Cripps, 1995)
- 2 **index theory**: the construction goes through for equilibria of **index** -1

Crash course in index theory

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- $J(\nu)$ is the Jacobian of all these, a $d \times d$ matrix, $d = \sum_i (m_i - 1)$, with **zero diagonal blocks**

$$J = \begin{pmatrix} 0 & * & * \\ * & 0 & * \\ * & * & 0 \end{pmatrix} \quad (\text{own indifference does not depend on own mixing})$$

Properties

- regular $\Rightarrow \det J \neq 0 \Rightarrow \text{index} = \pm 1$; pure NE $\Rightarrow \text{index} = +1$ (0×0 determinant)
- index is preserved if we drop actions outside the support
- dropping them and reversing the sign of u : $\text{index}_{-u}(\nu) = (-1)^d \text{index}_u(\nu)$

Crash course in index theory

For a regular NE ν ,

$$\text{index}_u(\nu) = \text{sgn det } J(\nu), \quad \text{where } J \text{ is the Jacobian of the indifference equations}$$

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Index theorem (Gül, Pearce, and Stacchetti, 1993; Ritzberger, 1994; Govindan and Wilson, 1997; DeMichelis and Germano, 2000)

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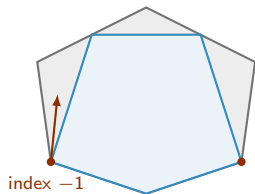
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- $\geq 2 \text{ NE} \Rightarrow$ some equilibrium has index -1
- if ν has index -1 , then ν is mixed and there is another equilibrium $\nu' \neq \nu$

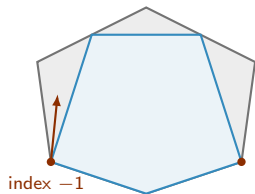
Constructing a CE outside the hull



≥ 2 equilibria, at most two mixers everywhere:

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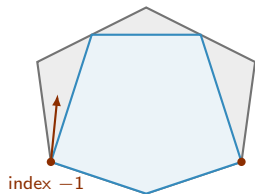
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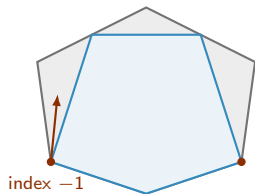
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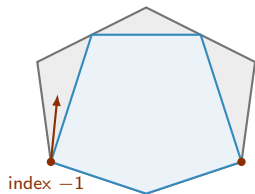
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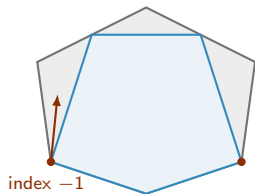
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Define,

$$\mu = \frac{\nu - \varepsilon \nu'}{1 - \varepsilon}$$

with $\varepsilon > 0$ small enough so that $\mu \geq 0$ and all outside-support constraints remain slack.

Then $\mu \in \text{CE} \setminus \text{conv}(\text{NE})$

► Why outside

When do the two approaches overlap?

Remark

A version of the index-based construction works at any equilibrium ν with ≥ 3 mixers, as long as

$$(-1)^{\sum_i (m_i - 1)} \cdot \text{index}(\nu) = -1$$

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A puzzle: is the parity-index interplay the essence of the problem, or an artifact of the construction?

A unique CE

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- but we can say a lot about its **coarse** analogue, CCE

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Plan

- ① characterize unique CCE ([Sandomirskiy and Wincelberg, 2026](#))
- ② outline what changes for unique CE (in progress)

CCE (Hannan, 1957; Moulin and Vial, 1978)

$\mu \in \Delta(A)$ is a CCE if for every player i and every ex-ante deviation $a'_i \in A_i$

$$\mathbb{E}_\mu[u_i(a)] \geq \mathbb{E}_\mu[u_i(a'_i, a_{-i})]$$

- the deviation is **before** seeing the recommendation $\Rightarrow \text{CCE} \supseteq \text{CE}$
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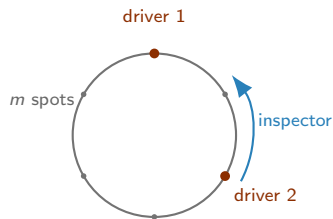
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What **else**?

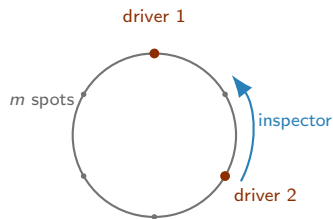
Intuition: a parking game

- 2 drivers: **park illegally** at one of m spots on a loop, or **park legally** (cost c)
- exactly one illegal $\Rightarrow$ caught for sure (ticket $t > c$)
- both illegal $\Rightarrow$ inspector starts at a uniformly random point, goes counter-clockwise; only the first found is ticketed



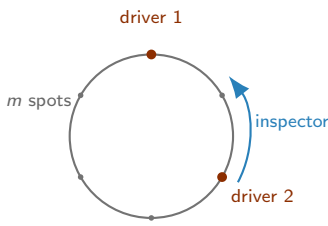
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- The diagram illustrates a circular parking lot with m spots, represented by small grey dots on a circle. Two drivers are shown: driver 1 is at the top spot, and driver 2 is at a spot clockwise from driver 1. An inspector is shown as a blue arrow moving counter-clockwise from a random starting point on the circle.
- a violator wants to park right after the other $\Rightarrow$ caught with probability only $1/m$
 - so **legal** is strictly dominant only when $\frac{t}{m} > c$
 - yet **legal** is the **unique CCE** already when $\frac{t}{2} > c$
 - **why:** under any joint violation at least one driver is caught with probability $\geq \frac{1}{2} \Rightarrow$ wants to deviate to **legal** ex-ante

Enforcement games and unique CCE

A game v is an **enforcement game** if there is $a^* \in A$ such that

$$v_i(a_i^*, a_{-i}) = 0, \quad \forall i, a_{-i}, \quad \sum_i v_i(a) < 0 \quad \forall a \neq a^*.$$

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- strategic equivalence: $u_i(a_i, a_{-i}) = \gamma_i(v_i(a_i, a_{-i}) + \beta_i(a_{-i}))$, $\gamma_i > 0$
- **Proof idea**
 - enforcement $\Rightarrow$ unique CCE:
 $\mu(A \setminus \{a^*\}) > 0 \Rightarrow \text{expected welfare} < 0 \Rightarrow \mathbb{E}_\mu[u_i] < 0$ for some $i \Rightarrow i$ would deviate to a_i^*
 - converse: LP duality

Beyond pure uniqueness

Unique CCE can be mixed. Matching pennies: the unique NE is the unique CCE

	H	T
H	$(1, -1)$	$(-1, 1)$
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If $\text{CCE}(G)$ is a singleton, then either it is a **pure** NE, or **exactly two** players mix over two actions each and the induced 2×2 game is of **matching-pennies type**

- in symmetric games, uniqueness $\Rightarrow$ purity
- idea: a unique CCE must be an **extreme point**; richer mixing destroys extremality

► Idea

From CCE to CE uniqueness

Rudov, Sandomirskiy, and Yariv (2025)

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≤ 2 mixers $\Rightarrow$ the unique CE is **pure**, or two players mix

we focus on the **pure** case

A pure unique CE (a preliminary result)

- CCE: each player enforces the target payoff by a **single-shot** deviation to a_i^*
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- the gap CE vs CCE: **one-shot** vs **multistage** chains
- future: Economic interpretation? Interesting examples?

Conclusion

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Economic takeaway

- mediation, communication, and learning bring new outcomes relative to decentralized play whenever it leaves anything to coordinate on

Nerdy takeaway

- a nonlinear object, the NE set and its index, governs the local structure of a linear object, the CE polytope
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Thank you!

Why ≥ 3 mixers fails extremality?

Support bound: i is subject to $|A_i|(|A_i| - 1)$ constraints. By [Winkler \(1988\)](#), an extreme CE has

$$\# \text{supp}(\mu) \leq 1 + \sum_i |A_i|(|A_i| - 1)$$

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Binary actions: $\leq 2n$ constraints $\Rightarrow$ extreme CE support $\leq 2n + 1$. A NE with k mixers has support 2^k ; dropping non-mixers,

$$2^k \leq 2k + 1 \quad \implies \quad k \leq 2$$

$\Rightarrow$ a NE with ≥ 3 mixers is not extreme

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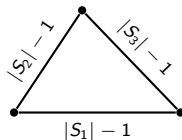
$\Rightarrow$ a NE with ≥ 3 mixers is not extreme

General supports: the polygon inequality. For any NE that is an extreme point of its Nash component, support sizes satisfy

$$|S_i| - 1 \leq \sum_{j \neq i} (|S_j| - 1) \quad \forall i$$

- each extra own action is one indifference constraint on opponents
 $\Rightarrow$ total constraints $\leq$ opponents' degrees of freedom
- Incompatible with the support bound once ≥ 3 mix

► Back



triangle inequality ($n = 3$)

Appendix: why the anti-Nash perturbation escapes the Nash hull

Consider $\mu = \frac{\nu - \varepsilon \nu'}{1 - \varepsilon}$, where ν' the anti-NE with $\text{supp}(\nu')$ strictly inside $\text{supp}(\nu)$

- $\Rightarrow \nu'$ omits some action $c_i \in \text{supp}(\nu_i)$ for a player $i \in \{1, 2\}$
- hence the player- i marginal increases at c_i :

$$\mu_i(c_i) = \frac{\nu_i(c_i)}{1 - \varepsilon} > \nu_i(c_i)$$

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- suppose for contradiction $\mu = \sum_{\ell} \lambda_{\ell} \nu^{\ell}$ with $\nu^{\ell} \in \text{NE}$, $\lambda_{\ell} > 0 \Rightarrow \text{supp}(\nu^{\ell}) \subseteq \text{supp}(\nu)$
- at μ , conditional on c_i , player i is indifferent across $\text{supp}(\nu_i)$
- $\Rightarrow$ the same holds under each $\nu^{\ell} \Rightarrow \text{supp}(\nu_i^{\ell}) = \text{supp}(\nu_i) \Rightarrow \text{supp}(\nu^{\ell}) = \text{supp}(\nu)$ for all ℓ
- in two-player games, no regular NE can share support with another NE, so every $\nu^{\ell} = \nu$
- then $\mu_i(c_i) = \nu_i(c_i)$, contradiction

no such decomposition exists $\Rightarrow \mu \in \text{CE} \setminus \text{conv}(\text{NE})$

Corollaries: classes with a unique CCE

- **two-player contests** with a strict pure NE and $p_1 + p_2 \equiv 1$
- **socially concave games** (Even-Dar, Mansour, and Nadav, 2009; Hart and Mas-Colell, 2015):
 A_i convex compact, u_i concave in a_i , convex in a_{-i} , welfare $\sum_i u_i$ strictly concave
 - ▶ includes linear Cournot and Tullock contests
- **potential games** with concave potential P and bounded cross-effects: at $a^* = \arg \max P$,

$$\text{diag}(H) + 2\text{offdiag}(H) \preceq 0, \quad H = \text{Hessian of } P$$

a limit on strategic complementarity/substitutability is necessary for CCE uniqueness

Appendix: a unique CCE must be “almost pure”

- $\text{CCE}(G)$ is a polytope; a singleton $\{\nu\}$ is (trivially) an **extreme point**, and ν is a quasi-strict NE
- a quasi-strict mixed NE with support sizes $k_1, \dots, k_m$ has

$$\# \text{ support} = \prod_i k_i, \quad \# \text{ binding constraints} \leq 1 + \sum_i k_i$$

- extreme $\Rightarrow \prod_i k_i \leq 1 + \sum_i k_i$
- integer solutions: **pure**, or **two players mixing over two actions** (2×2)

quasi-strict NE extreme in CCE $\iff$ pure or 2×2 mixing

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